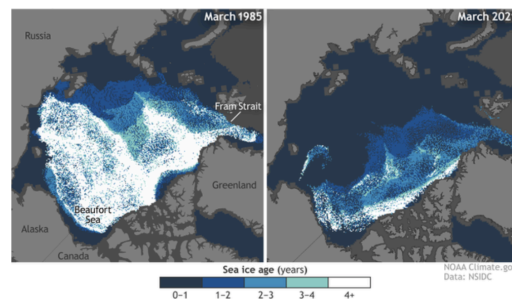


Arctic Sea Ice Loss

A DS 4002 Case Study by Kendall Freese



Background:

You are an intern at the University of Toronto who is fascinated with the Arctic. Its animals, its climate, and most especially, its ice. The Arctic serves as a global climate regulator. Ice reflects sunlight into space, helping to regulate global temperatures [1]. The more ice in the Arctic, the more regulated our planet. When ice melts and its overall mass decreases over time, the ocean water absorbs more heat from the sun. Heat that would've been prevented if there were substantial ice. You are rightfully concerned about ice loss in the Arctic, and alongside your studies in environmental science, you have kept tabs and followed research about the topic.

Context:

In your Glaciers, Ice, and Climate class, your professor, a renowned glaciologist, announces that he will embark on a month-long trip to the Arctic to collect ice samples, analyze melting patterns, and collaborate with glaciologists around the world. He will be bringing one student along on his trip to serve as an intern and help with research and data collection. To evaluate students objectively, the professor has created an application for the intern position. This application includes a 2-page essay on why you care about the Arctic sea ice loss problem and a project analyzing current ice trends in the Arctic. He is looking for meaningful findings and genuine interest. This is an opportunity of a lifetime for you.

Deliverable:

You want to submit an application that will stun the professor and guarantee you a spot on the trip. For the project, you decide that you want to analyze historical Arctic sea ice time series data, note key findings, and somehow predict near-zero (when the Arctic is considered ice-free). After extensive research, you clarify your goal of assessing seasonality, acceleration, volatility, and average decade decline in Arctic sea ice and developing a predictive algorithm that forecasts future ice loss and estimates when levels will reach near-zero, targeting 80% observed values within the 80% prediction interval. You will build three models—linear, quadratic, and ARIMA—to accomplish these objectives and ultimately will determine the best model for calculating near-zero. You will conduct your analysis, summarize the findings and methodology in your application, and apply for the internship position.

[1]"Arctic sea ice maximum: Crisis as ice melts and risks rise - WWF Arctic," *WWF Arctic*, Mar. 28, 2025.
<https://www.arcticwwf.org/newsroom/features/arctic-sea-ice-maximum-crisis-as-ice-melts-and-risks-rise/>